Donavalli Sai Durga Rishi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.23
Narayana Junior College	Class XII (APBIE)	2023	98.00%
Kennedy High School	Class X (SSC)	2021	99.67%

SCHOLASTIC ACHIEVEMENTS

- Secured Overall Rank 11 in The Integral Cup 2026 Season 1 Finals sponsored by Optiver, DaVinci Trading.

- Secured Overall Rank 110(India) in the annual worldwide IMC Trading Competition Prosperity 4.

- Secured 1st Position in Round 1 of The Integral Cup 2025, VGS at IITH sponsored by Jane Street, Optiver, QRT.

- Secured 1st Position in both Round 1 and 2 of Waveform Wars (Elan & nvision with Australian Government).

- Secured 27th Rank in Round 1 and 14th Rank in Round 2 of the Srinivasa Ramanuja Maths Competition, hosted by IIT Madras and IIT Bombay Mathematics Clubs and sponsored by Jane Street.

- Secured 10th Position in Math Bowl part of the Inter IIT Tech Meet 14.0 representing IIT Hyderabad.

- Achieved gold level in the World Quant Brain Challenge with more than 10k+ points by submitting alpha strategies.

- Qualified to stage 2 of the International Quant Championship held by WorldQuant BRAIN.

WORK EXPERIENCE

Internship | Google (via Uber) **June 2026 - Present**

- Working with a team affiliated with Google DeepMind on benchmarking multiple Gemini models, designing long-horizon data science tasks to evaluate multi-step reasoning and end-to-end problem solving.

- Worked on document generation and parsing tasks, then developed long-horizon finance tasks requiring sequential data analysis, modeling and verification across realistic multi-stage workflows.

- Authored golden reference solutions for NP-complete combinatorial problems and mixed-integer linear programs (MILP), used as ground truth to rigorously assess Gemini models reasoning accuracy.

PROJECTS

Topological Regime Detection for High-Frequency Market Making | Self Research Project

- Proposed a regime-detection statistic for order book microstructure, embedding rolling LOBSTER Level-3 windows into truncated path-signature space and scoring them via Fisher Information trace.

- Implemented truncated signatures in pure PyTorch via Chen's identity with test-verified correctness, and estimated Fisher trace from per-sample scores using vmap(grad) over signature features.

- Designed an online Welford-calibrated threshold test with day-boundary resets to avoid lookahead; on synthetic data with injected toxic-flow bursts, it fired only inside the injected regimes.

Symbolic Neural ODEs for Irregular Time Series | Research Project under Prof. Srijith P.K, IIT Hyderabad

- Developing symbolic neural ODE models for irregularly sampled time series, learning continuous-time dynamics directly from real-world temporal data with missing and unevenly spaced observations.

- Designing adaptive step-size solvers that handle missing observations and uneven sampling intervals, allowing learned dynamics to be integrated across arbitrary gaps between consecutive observations.

- Proposing interpretable symbolic representations of learned dynamics and evaluating predictive performance and

interpretability on real-world datasets with irregular sampling patterns.

Shortcut Learning Analysis in Vision Models | Self Research Project

- Studying shortcut learning in deep networks by constructing controlled datasets with background and texture biases to expose model reliance on spurious correlations over semantic features.

- Designed perturbation-based experiments and representation probing techniques to analyze how robustness degrades under distribution shifts when models latch onto dataset artifacts.

- Evaluated mitigation strategies including stronger augmentations, feature regularization and architecture modifications, measuring how effectively each reduces reliance on shortcut features.

Hardware Profiling Aware Compression for Language Models | Self Research Project

- Developing compression strategies for transformer-based language models guided by hardware profiling metrics such as latency, memory bandwidth and compute utilization instead of parameter counts.

- Implemented structured pruning, weight sparsification and quantization techniques to reduce inference cost of large language models while preserving predictive performance for deployment.

- Built benchmarking pipelines with profiling tools to measure real hardware speedups across CPU and GPU environments, analyzing trade-offs between compression ratio, throughput and model accuracy.

Explanation Consistency in Vision Transformers | Self Research Project

- Investigating robustness of explanation methods for Vision Transformers by enforcing consistency of attributions across stochastic perturbations and input augmentations to test their reliability.

- Implemented evaluation pipelines comparing attribution stability across gradient, attention and perturbation-based methods, including Integrated Gradients, GradCAM and attention rollout.

- Designed metrics quantifying explanation variance across stochastic forward passes and augmentation distributions, running large-scale experiments in PyTorch and HuggingFace on benchmark datasets.

Inter IIT Tech Meet 13.0 - IIT Bombay

- Designed a deepfake detection pipeline using Binary Neural Networks (BNNs) with uncertainty estimation for robust inference on synthetic media, targeting efficient and reliable classification.

- Built FFT-based preprocessing to extract frequency-domain representations, capturing compression artifacts and generative model signatures and localizing noise patterns for interpretability.

- Evaluated model robustness across multiple compression levels and image perturbations, testing whether synthetic media remained reliably detectable under realistic degradation conditions.

Bharatiya Antariksh Hackathon - ISRO

- Built a GeoAI Assistant translating natural-language geospatial queries into executable GIS workflows, integrating live datasets with GeoPandas, Rasterio and PyQGIS for adaptive spatial analytics.

- Implemented LLM-based multi-agent orchestration with NER for task decomposition, breaking complex geospatial requests into sequential data-processing and analysis steps for execution.

- Designed an interactive, feedback-driven dashboard enabling explainable visualizations of spatial analysis results, so users could inspect and refine each step of the generated workflow.

Bounding Box Optimization for Rooftop Detection

- Augmented YOLOv8 with RNN-based refinement modules for post-prediction bounding box correction, improving localization accuracy for rooftop detection on noisy satellite imagery.

- Integrated temporal cues and neighborhood context into the refinement stage and designed post-prediction regression correction algorithms to reduce bounding box localization error.

- Evaluated segmentation accuracy improvements across multiple rooftop geometries and environmental conditions, validating the refinement's gains on diverse real-world satellite scenes.

Job Portal - DBMS Project

- Created a full-stack job portal supporting user authentication, resume uploads and job discovery for seekers, with advanced filtering and search functionality across all job listings.

- Designed a normalized MySQL schema using joins, indexing and integrity constraints to reduce redundancy, and optimized SQL queries to keep filtering and search fast and scalable.

- Built administrative dashboards enabling employers to manage job postings and monitor application statistics, giving

recruiters a clear view of applicant activity per listing.

RISC-V Instruction Set Simulator - Computer Architecture Project

- Developed a RISC-V instruction-level simulator in C supporting instruction decoding, memory access, register updates and control flow execution following the RISC-V ISA specification.

- Implemented arithmetic, logical, load/store and branch instructions, and built execution pipeline logic to simulate instruction flow and program control behavior across test programs.

- Validated simulator correctness using extensive custom test cases and instruction trace comparisons, ensuring register and memory states matched expected outputs after each instruction.

American Express Decision Track Challenge - Unstop

- Built a ranking model for customer offer-acceptance prediction on large-scale tabular data, engineering click-through-rate and time-based behavioral features to improve ranking accuracy.

- Trained gradient boosted decision tree models using LightGBM optimized for the MAP@7 evaluation metric, and tuned hyperparameters for consistent, stable leaderboard performance.

- Applied SHAP analysis to interpret feature importance and identify key drivers of offer acceptance, and studied feature interactions and correlations to guide feature selection.

POSITIONS OF RESPONSIBILITY

- Science & Technology Secretary for the tenure 2025-26, Students Gymkhana, IIT Hyderabad.

- Hospitality Deputy Coordinator, Elan & nvision (2023-24)| Core Member, Lambda (Software Development Club) (2024-25).

- Teaching Assistant for the 3 credit course Maths for EE under Prof. Aditya Siripuram (Aug 2024-Nov 2024).

- Teaching Assistant for the 3 credit course Matrix Theory under Prof. KR Mopuri (Aug 2026-Nov 2026).

SKILLS

- Languages: C/C++, Go, Python, SQL, HTML, CSS, JavaScript, R

- Technologies: Firebase, PostgreSQL, Docker, MongoDB, Git/GitHub/GitLab, Docker

- Frameworks: React, Next.js, Vite, ExpressJS, Django, TensorFlow, PyTorch, Hugging Face, Socket.IO, Bootstrap, Tailwind CSS, Vercel, Render, Scikit-Learn, Numpy, Pandas, Matplotlib, Solid Edge

- Domains: Web/App/Software Development, AI/ML Solution Development

RELEVANT COURSES

Probability and Random Variables, Introduction to Programming, Programming for AI, Matrix Theory, Applied Statistics, ODE, Foundations of Machine Learning, Algorithms, DSA, Computer Architecture, DBMS, OS-1, Compilers-1 , Deep Learning, Computer Vision, NLP, AI in Finance, Multi Agent Systems

EXTRACURRICULAR

- Silver Medal in Football, DIESTA'25, IITH

- Part of the Runner Up Team in the Institute Water Polo League, IIT Hyderabad.